

# AgriGenAI: The Crop Saver

**Multi-Agent System using Langflow & IBM Watsonx.ai**

**Project on Agriculture domain: " AgriGenAI: The Crop Saver"** (Multi-Agent solution)

**The Challenge: -**

Farmers often spot symptoms on their crops (e.g., "yellow leaves," "black spots") but struggle to distinguish between fungal infections, nutrient deficiencies, or pests. Misdiagnosing leads to spraying the wrong chemicals, wasting money and harming the soil.

**The Objective**

Build a **Multi-Agent AI system** that helps farmers accurately diagnose crop issues based on symptoms and provides actionable treatment recommendations. The system reduces misdiagnosis, prevents unnecessary chemical usage, and promotes sustainable farming practices.

**Key Features**

## 1. Symptom-Based Diagnosis

Identifies crop problems from inputs like “yellow leaves” or “black spots”

Differentiates between diseases, pests, and nutrient deficiencies

**Multi-Agent Architecture**

Diagnosis Agent → Identifies the issue

Treatment Agent → Suggests remedies

Advisory Agent → Provides preventive guidance

Smart Treatment Recommendations

Suggests correct pesticides, fertilizers, or organic solutions

Avoids overuse of harmful chemicals

Recommends cost-effective options

Preventive Care Guidance

Provides tips to prevent future crop issues

Suggests best farming and soil practices

**Context-Aware Suggestions**

Considers crop type, region, and environmental conditions

Ensures practical and relevant advice

## **Farmer-Friendly Support**

Uses simple, easy-to-understand language

Improves decision-making and reduces guesswork

### **Prerequisites**

- Watsonx API Key from IBM Cloud
- Watsonx Project ID from IBM Cloud
- Watsonx API Endpoint for Dallas: <https://au-sydney.ml.cloud.ibm.com>

### **Tools & Technologies Used:**

- IBM Cloud

- Langflows